

Empowering biomedical evidence exploration and synthesis with deep knowledge graph research

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Zifeng Wang¹✉, Zheng Chen², Ziwei Yang³, Xuan Wang², Qiao Jin⁴, Yifan Peng⁵, Zhiyong Lu⁴ & Jimeng Sun^{1,6}✉

Compared with generic artificial intelligence agents, deep research agents perform longer-horizon reasoning and deeper literature exploration to investigate complex questions. Here we present DeepEvidence, a deep research agent for evidence exploration and synthesis across heterogeneous biomedical knowledge sources. DeepEvidence advances deep research through coordinated multi-agent collaboration combining breadth-first and depth-first research strategies to search, explore and aggregate evidence from multiple biomedical knowledge bases and literature. It also incrementally constructs an evidence graph of key entities and observations to support transparent tracking, attribution and validation of the research process. DeepEvidence substantially outperforms generic artificial intelligence agents across four open benchmarks. We further establish seven benchmark tasks spanning major stages of biomedical discovery, including drug discovery, preclinical experimentation, clinical trial development and evidence-based medicine. DeepEvidence demonstrates substantial improvements in systematic evidence exploration and synthesis. These results highlight the potential of deep research agents to accelerate biomedical discovery and translational research.

Literature and knowledge bases are fundamental to biomedical discovery, such as for literature mining, chemical and drug annotation, gene and protein characterization, pathway analysis and disease-phenotype understanding. Prominent examples include PubTator¹ and PKG² for publication-derived knowledge, Kyoto Encyclopedia of Genes and Genomes (KEGG)³ and Gene Ontology⁴ for pathways and functional annotation, OpenTargets⁵ for gene-disease associations, and BioThings⁶ for unified chemical and gene annotations. Together, these sources provide rich, complementary representations of biomedical entities, relations and annotations that underpin modern research workflows. Yet, leveraging them collectively remains highly

challenging: they differ markedly in structure, evolve continuously and rarely align seamlessly, resulting in limited interoperability and cross-resource reasoning difficult in practice.

Artificial intelligence (AI) agents have been adopted for systematic literature review^{7,8}, genomics research^{9,10}, spatial biology¹¹ and biomarker discovery¹², and have been developed as AI scientists that interact with diverse scientific knowledge sources and computational tools to streamline, enhance, or automate components of the research process^{13–15}, including ones adapted specifically to biomedical tasks^{16,17}. In parallel, deep research agents are proposed as a special AI agent that can autonomously search, reason and synthesize evidence from the

¹Keiji AI, Seattle, WA, USA. ²Institute of Scientific and Industrial Research, Osaka University, Osaka, Japan. ³Bioinformatics Center, Institute for Chemical Research, Kyoto University, Kyoto, Japan. ⁴Division of Intramural Research, National Library of Medicine, National Institutes of Health, Bethesda, MD, USA. ⁵Department of Population Health Sciences, Weill Cornell Medicine, New York, NY, USA. ⁶School of Computing and Data Science, University of Illinois Urbana-Champaign, Urbana, IL, USA. ✉e-mail: zifeng@keiji.ai; jimeng@illinois.edu

literature, as exemplified by OpenAI DeepResearch¹⁸, Gemini Deep Research¹⁹ and Perplexity Deep Research²⁰. However, generic deep research agents primarily operate over unstructured web text and are not well suited for biomedical research, where critical evidence is distributed across heterogeneous resources, including biomedical literature, structured databases, ontologies and knowledge graphs. Effectively bridging these diverse knowledge sources to uncover subtle relationships and connections remains a key challenge for systematic biomedical evidence exploration and synthesis.

Here, we present DeepEvidence, an AI agent system specifically designed to enable deep research across heterogeneous biomedical knowledge sources. Unlike generic deep research systems, DeepEvidence leverages specialized biomedical tooling to locate and link entities across heterogeneous knowledge bases and literature sources, and to navigate them through a coordinated combination of breadth-first research (BFRS) and depth-first research (DFRS): two complementary strategies for systematic evidence exploration. DeepEvidence further incorporates an internal evidence-graph memory that is continuously updated as the agent traverses and synthesizes information, providing a structured and transparent record of the discovery process. We instantiate these principles in an orchestrator–subagent architecture comprising dedicated BFRS and DFRS agents that collaborate to build and refine the final evidence graph (Fig. 1).

Our empirical evaluation begins by assessing DeepEvidence's performance across established biomedical benchmarks that require deep exploration and reasoning over literature, including Humanity's Last Exam (HLE)²¹, LabBench-LitQA2²², SuperGPQA²³ and TrialPanorama-EvidenceQA²⁴. We then demonstrate DeepEvidence's practical utility across four critical stages of the biomedical discovery lifecycle: (1) drug discovery, (2) preclinical in vivo experimentation, (3) clinical trial development and (4) evidence-based medicine (EBM). Together, these results illustrate the potential of knowledge-graph-driven deep research to support and accelerate biomedical discovery.

Overview of DeepEvidence

DeepEvidence is a multi-agent framework with an orchestrator agent that delegates tasks to a set of specialized subagents, called research agents here (Fig. 1a). The orchestrator also maintains an internal evidence graph that records the key entities and relations discovered during the research process (Fig. 1b). There are two types of research agent: BFRS and DFRS agents. BFRS performs multigraph queries to rapidly survey first-hop neighbourhoods (for example, identifying drugs linked to a disease across disease–drug or gene–disease graphs). DFRS builds on the key entities surfaced by BFRS and performs deeper, multihop exploration (for example, recursively tracing references across publications to construct a comprehensive evidence graph for systematic literature review). We refer to these two modes as BFRS and DFRS, rather than the conventional BFS/DFS, because the proposed AI agents choose traversal steps autonomously rather than follow fixed graph-theoretic rules. Also, AI agents can bridge across heterogeneous knowledge graphs by recognizing shared entities or inferring cross-graph relations (referred to as bridge entities in Fig. 1a).

We created an environment allowing DeepEvidence to explore diverse biomedical sources at scale (Fig. 1c). Many biomedical sources are accessible through heterogeneous web application programming interfaces (APIs), so to support efficient BFRS, we implemented unified entity-search and data-fetching interfaces across major biomedical domains, including publications, clinical trials, chemicals, genes, drugs, diseases, phenotypes and proteins. For example, a single drug query issued by an agent can automatically trigger parallel lookups across KEGG³, ChEMBL²⁵, OpenFDA²⁶, OpenTargets⁵ and other resources. Because these operations involve large numbers of API calls and produce heterogeneous data, we equipped the system with an execution sandbox that enables agents to conduct research

programmatically. Within this sandbox, agents can write and execute code to query knowledge-graph APIs at scale, persist retrieved results as structured files (for example, spreadsheets) and perform downstream extraction and analysis all within a controlled environment.

DeepEvidence takes the input research task, then has the orchestrator determine the research plan first (Fig. 1d). At the outset, the orchestrator selects the sequence and interplay of BFRS and DFRS and identifies the research targets for each step. Each research agent then writes programs to query knowledge sources and then extract and analyse the retrieved data. The orchestrator then concludes with the findings aggregated from research agents' outputs.

DeepEvidence excels on biomedical research benchmarks

We evaluated DeepEvidence on four challenging open benchmarks that require general biomedical knowledge, multihop reasoning and evidence-based research: HLE²¹, LabBench²², SuperGPQA²³ and TrialPanorama²⁴ (Fig. 1e). To highlight the contribution of our proposed Deep Research framework, we compared DeepEvidence against strong baselines: (1) two frontier large language models (LLMs) (GPT-5 and Sonnet-4.5) equipped with literature-search capabilities, and (2) two general-purpose biomedical agents (Biomni¹⁶ and ToolUniverse¹⁷). For fairness, all agent systems, including DeepEvidence, were run with the same underlying LLM (GPT-5).

Across all four benchmarks, DeepEvidence consistently delivered the strongest performance (Fig. 1e). On HLE-Medicine, DeepEvidence achieved 40.0%, substantially outperforming Biomni (20.0%), ToolUniverse (10.0%), Sonnet-4.5 (3.3%) and GPT-5 (3.3%). On LabBench-LitQA2, a benchmark requiring retrieval and synthesis of publication-derived evidence, DeepEvidence achieves 64.7% accuracy that substantially outperform Biomni (48.9%) and ToolUniverse (25.0%). On SuperGPQA-Medicine-Hard, which emphasizes difficult mechanistic reasoning, DeepEvidence achieved 47.1%, exceeding all baselines (Biomni 40.7%; Sonnet-4.5 43.6%). Finally, on TrialPanorama-EvidenceQA, a clinical-evidence reasoning task grounded in real-world clinical trials, DeepEvidence obtained 96.0%, outperforming both Biomni (84.0%) and Sonnet-4.5 (88.0%).

These results demonstrate the value of deep knowledge graph research in biomedicine. Although agents like Biomni and ToolUniverse have access to many knowledge graph APIs, they typically perform shallow exploration, issuing only a few searches before answering. Their large tool suites can also introduce tool confusion, leading agents to select irrelevant tools or follow incorrect reasoning paths. By contrast, DeepEvidence carries out structured BFRS–DFRS exploration guided by a continually updated evidence graph, enabling deeper, more targeted and more reliable knowledge graph investigation. This design directly supports superior performance on tasks that require multihop reasoning and evidence synthesis.

To further assess DeepEvidence's generalization in realistic biomedical research settings, we curated six new benchmark tasks that span the full lifecycle of biomedical discovery, including drug discovery, preclinical research, clinical trial development and EBM. We introduce and analyse each benchmark in detail in the following sections.

DeepEvidence for drug discovery and preclinical research

Drug discovery and preclinical research constitute the first two stages of the drug development lifecycle. Drug discovery focuses on identifying actionable biological targets, while preclinical research aims to elucidate mechanisms of action and evaluate on-target biological responses in experimental models. In this section, we assess the capability of DeepEvidence to support both stages. We evaluate DeepEvidence on a target identification task for drug discovery, followed by two preclinical mechanistic reasoning tasks that probe pathway-level interpretation and in vivo metabolic response prediction (Figs. 2a and 3).

The first task is target identification, which requires identifying promising therapeutic or diagnostic targets for a given disease context.

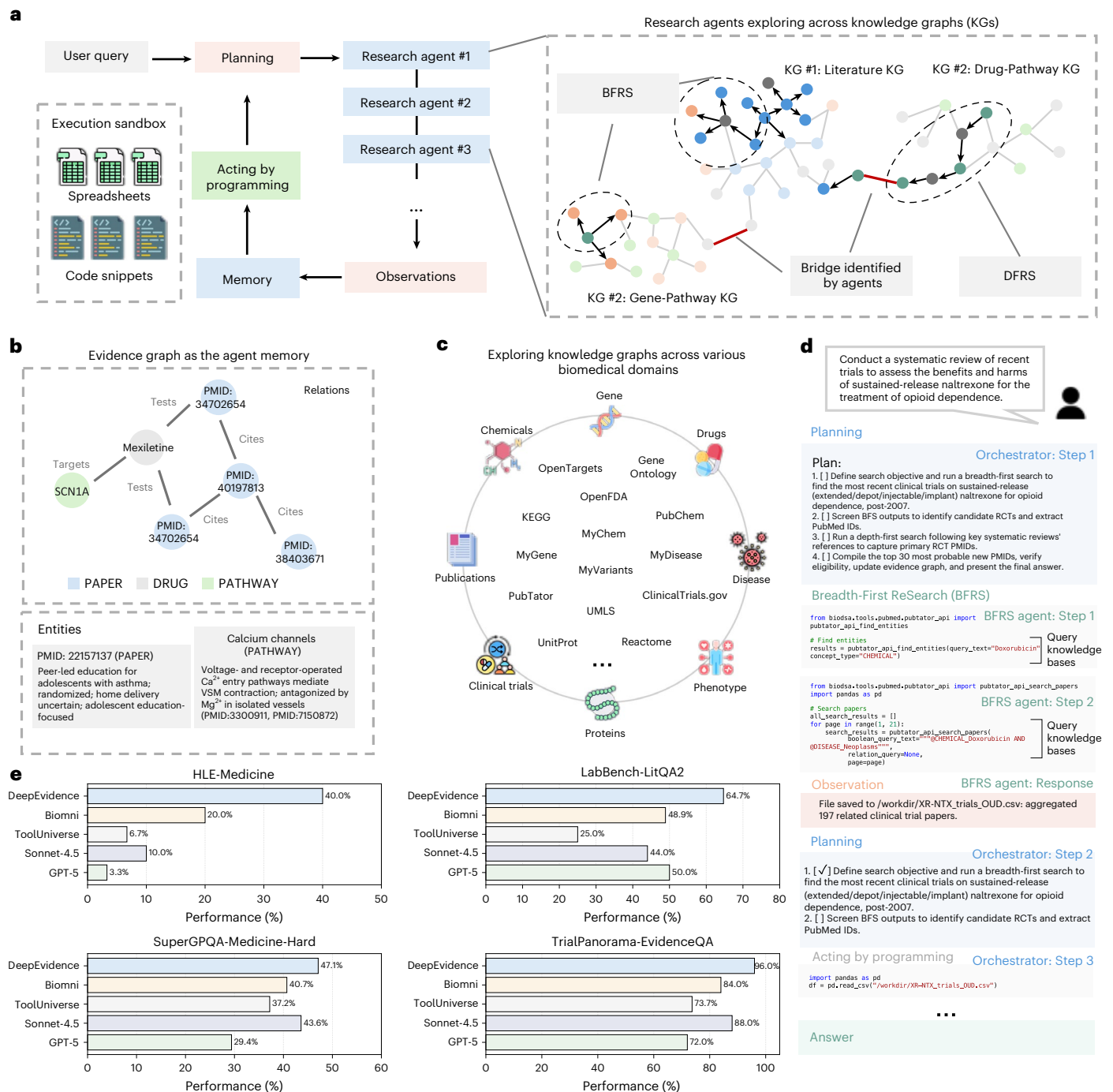


Fig. 1 | Overview of DeepEvidence and benchmark performance. a, System overview showing the orchestrator coordinating planning, action and memory updates across research agents. A user query is translated into a planned sequence of research agents that act within an execution sandbox and update a shared memory through observations. These agents perform coordinated breadth-first and depth-first exploration across multiple biomedical knowledge graphs to identify cross-graph links and build an integrated evidence view. **b**, The agent incrementally builds a structured graph that links papers, drugs, genes and pathways through relations such as targets, tests and citations. This

evidence graph serves as a persistent memory that organizes discovered entities and supports downstream reasoning. **c**, Overview of the diverse biomedical knowledge graphs that the agent can access across domains, including genes, diseases, drugs, chemicals, proteins, phenotypes, publications and clinical trials. **d**, A step-by-step example showing how the agent plans, queries knowledge bases, executes BFRS, collects observations and uses programming actions to complete a systematic review task. **e**, Benchmark results showing that DeepEvidence consistently outperforms strong baseline systems across four biomedical reasoning tasks. RCT, randomized clinical trial.

As illustrated in Fig. 2a, the agent is provided with a research question together with constraints on target purpose and methodological category, and must iteratively search, aggregate and evaluate evidence to converge on a well-grounded target. The performance is shown in Fig. 2b. On this task, DeepEvidence achieves an accuracy of 68%, substantially outperforming Biomni at 56%, ToolUniverse at 40% and

general-purpose LLMs at 42%. This gap primarily reflects differences in research strategy. Through inspection of agent trajectories, we observe that baseline methods such as Biomni and ToolUniverse often struggle to identify appropriate tools and effectively orchestrate the research process due to the large pool of candidate tools and noisy retrieval outputs. Moreover, the retrieved results frequently contain conflicting

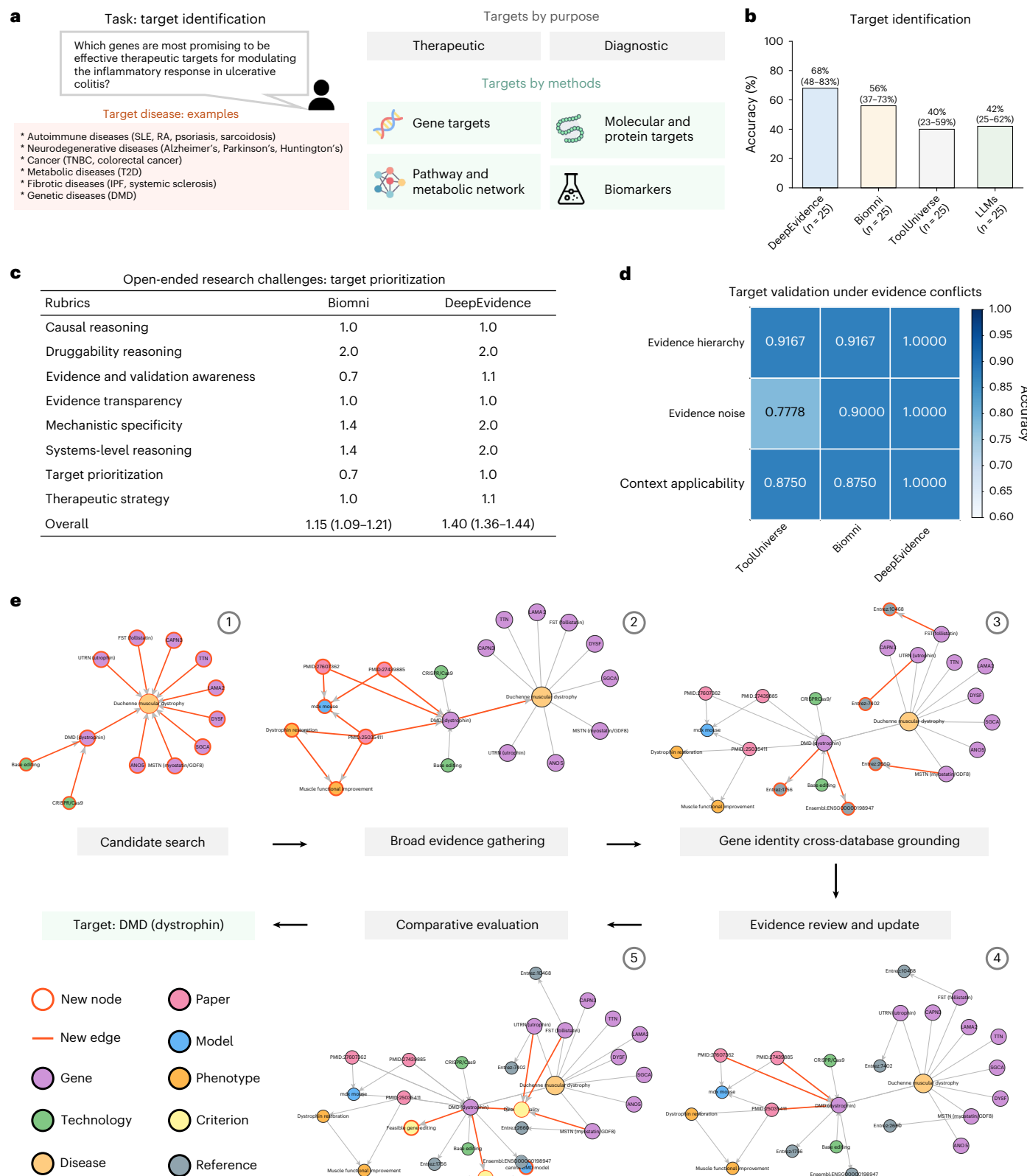


Fig. 2 | Experiment results of target identification. **a**, Illustration of a target identification task that defines the research question, disease scope, target purposes and methodological categories for therapeutic and diagnostic discovery. SLE, systemic lupus erythematosus; RA, rheumatoid arthritis; TNBC, triple-negative breast cancer; T2D, type 2 diabetes; IPF, idiopathic pulmonary fibrosis; DMD, duchenne muscular dystrophy. **b**, Accuracy comparison for the target identification task showing that DeepEvidence achieves higher performance than baseline systems across evaluation settings. All values are presented as mean

(95% confidence interval (CI)); the same format applies to all bar plots in this figure. **c**, Human evaluation results on open-ended target prioritization tasks. Bold values indicate that DeepEvidence outperforms the baseline. **d**, Performance on target validation under context-specific evidence conflicts. **e**, An example end-to-end target identification workflow, where the agent conducts candidate search, broad evidence gathering, gene identity grounding across databases, evidence review and update, and comparative evaluation, progressively constructing and refining an evidence graph to identify and validate a disease-associated gene target.

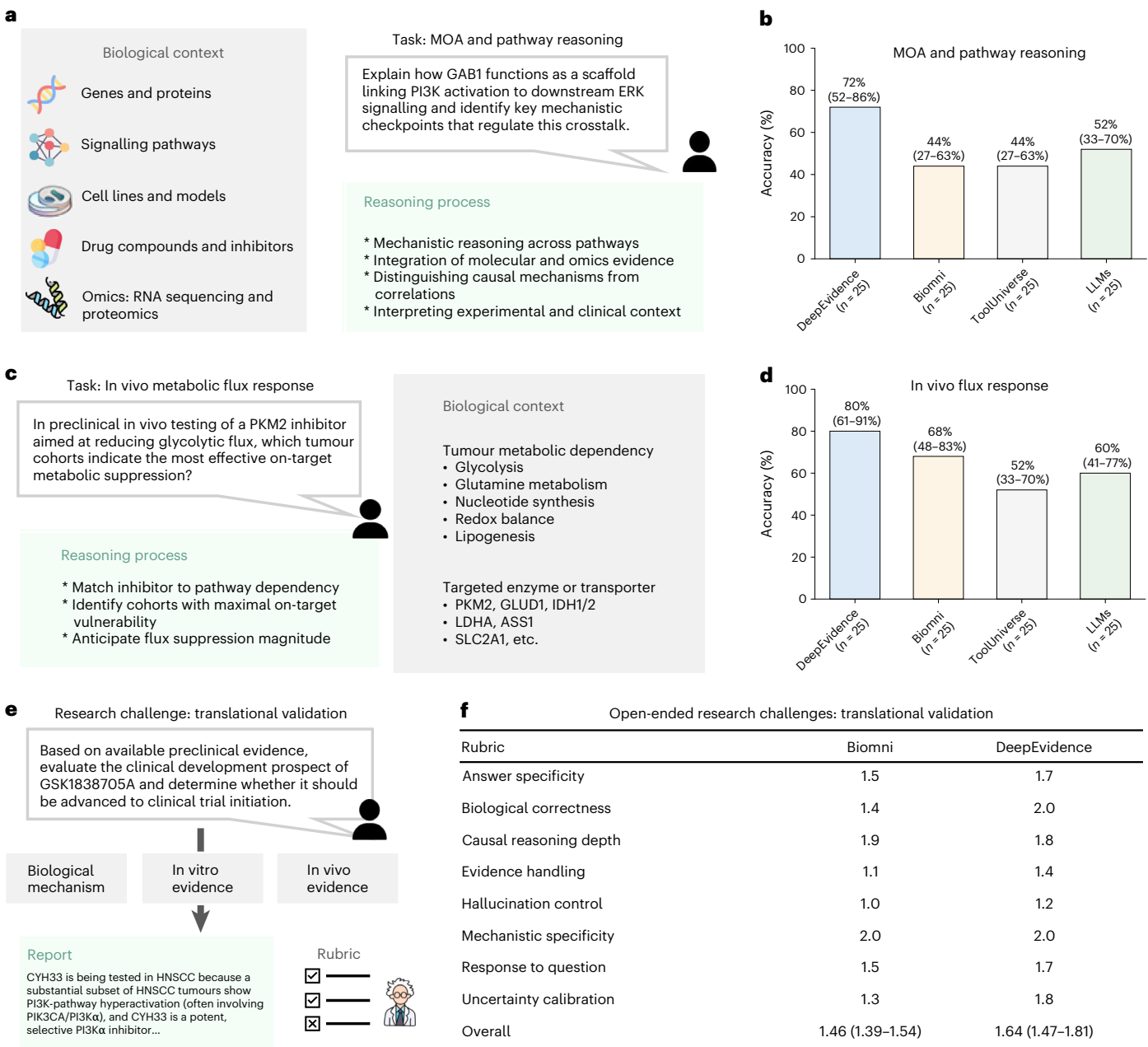


Fig. 3 | Experiment results of mechanistic reasoning tasks. **a**, Illustration of a mechanism of action and pathway reasoning task that specifies biological context, molecular entities and reasoning objectives to explain signalling crosstalk and identify key regulatory checkpoints. **b**, Accuracy comparison for the mechanism-of-action (MOA) and pathway reasoning task, showing that DeepEvidence achieves the highest performance compared with baseline systems. All values are presented as mean (95% CI); the same format applies to all bar plots in this figure. **c**, Illustration of an in vivo metabolic flux response task

where the agent reasons over tumour metabolic dependencies, target enzymes and pathway vulnerabilities to identify cohorts with effective on-target flux suppression. HNSCC, head and neck squamous cell carcinoma. **d**, Accuracy comparison for the in vivo flux response task showing that DeepEvidence consistently outperforms baseline methods across evaluation settings. **e**, Illustration of the open-ended translational validation task. **f**, Human evaluation results comparing Biomni and DeepEvidence on translational validation. Bold values indicate that DeepEvidence outperforms the baseline.

evidence and mixed contexts, creating challenges for downstream evidence aggregation and reasoning. By contrast, DeepEvidence performs more focused and structured exploration of biomedical literature and knowledge sources, enabling deeper and more comprehensive evidence synthesis. To further evaluate this capability, we introduce a new set of binary target validation challenges involving target candidates with conflicting evidence in the literature, including evidence hierarchy conflict, evidence noise and context applicability conflict (see the Methods for detailed definitions). These tasks require agents to retrieve, reconcile and reason over conflicting evidence before

reaching a final judgement. As shown in Fig. 2d, DeepEvidence achieves the strongest performance, suggesting that its broader evidence coverage and structured reasoning process better support the aggregation and interpretation of heterogeneous evidence without conflating context across sources. We have also provided an illustrative example of the evidence graph expansion process in Fig. 2e.

We further introduce open-ended research challenges that require free-form, end-to-end reasoning without predefined answer options. Target validation is formulated as validating and prioritizing hypotheses derived from data-driven discovery (for example, RNA sequencing

differential expression). In this task, agents are provided with a set of candidate genes and asked to design a prioritization strategy, identify high-confidence targets and justify their decisions based on biological mechanisms, druggability and supporting evidence. An example instance is presented in Supplementary Table 9. Outputs are evaluated by domain experts using structured rubrics covering causal reasoning, mechanistic specificity, prioritization quality, therapeutic strategy, druggability, evidence awareness and systems-level reasoning; detailed rubrics are provided in Supplementary Table 11. As shown in Fig. 2c, DeepEvidence outperforms Biomni in five of eight evaluation dimensions, with the remaining three being comparable. Overall, DeepEvidence achieves a substantially higher score of 1.40 compared with Biomni's 1.15 ($P < 0.001$).

We next assess DeepEvidence on preclinical research challenges spanning two major categories: (1) mechanism-of-action and pathway reasoning (Fig. 3a), and (2) in vivo metabolic flux response prediction (Fig. 3c). Representative task instances are provided in Supplementary Tables 3 and 4, respectively. For the first category, mechanism-of-action and pathway reasoning tasks require agents to analyse realistic preclinical scenarios such as explaining signalling crosstalk, interpreting drug mechanisms and biomarkers, analysing transcriptomic and phosphoproteomic data, and connecting molecular observations to phenotypic or therapeutic outcomes. As shown in Fig. 3b, DeepEvidence achieves an accuracy of 72%, substantially outperforming Biomni and ToolUniverse (both 44%) as well as general-purpose LLMs (52%). These results suggest that DeepEvidence more effectively integrates heterogeneous experimental evidence into coherent and biologically plausible mechanistic explanations, whereas baseline methods often struggle to reconcile multisource data or reason beyond surface-level associations. For the second category, the in vivo metabolic flux response task reflects common preclinical study designs where agents must identify tumour cohorts or biomarkers most likely to exhibit strong on-target metabolic suppression following enzyme inhibition. As illustrated in Fig. 3d, DeepEvidence achieves an accuracy of 80%, outperforming Biomni (68%), general-purpose LLMs (60%) and ToolUniverse (52%). We observe a similar trend in which baseline methods often fail to integrate metabolic context with experimental design constraints, while DeepEvidence more reliably aligns target enzymes with pathway-level dependencies through focused, evidence-grounded exploration.

We further evaluate the methods on translational validation tasks, an open-ended research challenge we curate in which experts manually assess agent output quality using structured rubrics (Fig. 3e). An example task instance is provided in Supplementary Table 10, and the evaluation rubrics are presented in Supplementary Table 12. The results are presented in Fig. 3f. Consistent with prior observations, DeepEvidence outperforms Biomni in six of eight evaluation dimensions, achieving a higher overall score of 1.64 compared with 1.46 ($P < 0.05$).

DeepEvidence for clinical trial development

Every new therapeutic candidate must undergo human studies through clinical trials to determine its safety and efficacy for a specific disease and patient cohort (Fig. 4a). In this section, we assess the capability of DeepEvidence to support clinical trial development, with a focus on core design decisions. Specifically, we developed three challenging tasks for trial planning: sample size estimation, drug regimen design and surrogate endpoint discovery. Successfully completing these tasks typically requires a broad survey of prior clinical evidence together with expert-level reasoning about disease biology and therapeutic mechanisms.

The first task is 'sample size estimation', which requires determining the number of participants needed to ensure that the study reaches the desired significance level with sufficient statistical power. As illustrated in Fig. 4b, the agents receive a question together with statistical assumptions and treatment group definitions, and then attempt to

infer the required sample size. The performance is shown in Fig. 4c. On this task, DeepEvidence reached an accuracy of 68%, substantially surpassing the three baselines: Biomni (20%), ToolUniverse (32%) and a PubMed-search LLM (24%). We attribute the superior performance of DeepEvidence to its ability to conduct a thorough review of sample size estimation strategies used in related studies, and form an overview of the sample size patterns and design set-ups from similar clinical trials. By contrast, the general-purpose agents either perform only shallow searches or attempt to infer the sample size directly without gathering substantive evidence.

The second task is 'drug regimen design' (Fig. 4d), which requires selecting an appropriate strategy for a drug combination, including starting dose levels, escalation approach, dose-limiting toxicity (DLT) definitions and treatment schedule choices that jointly balance therapeutic effect and participant safety. Each question requires a review of the historical evidence for a specific drug combination, and the agent must identify the correct regimen strategy based on this evidence and on established rubrics. The performance is shown in Fig. 4e. DeepEvidence achieved 52% in accuracy, outperforming Biomni (36%), ToolUniverse (28%) and PubMed-search LLMs (20%). Analysis of agent behaviour revealed that DeepEvidence consistently retrieved information from Food and Drug Administration-approved drug references and relevant literature, allowing it to form a comprehensive view of historical dosing and toxicity patterns, whereas the baseline agents often produced incomplete evidence gathering.

The third task is 'surrogate endpoint discovery' (Fig. 4f), in which the agent ranks the plausibility of candidate surrogate endpoints for a novel therapy based on its mechanism of action and biological context. Solving this task requires searching biomedical knowledge bases to identify disease pathways, drug targets and related biological processes, and then determining which endpoints are mechanistically supported. The results in Fig. 4g are evaluated using average precision, recall and F1 score. DeepEvidence achieved 73.2% in precision, 75.0% in recall and 73.3% in F1. Biomni achieved 60.8%, 64.3% and 60.9%. ToolUniverse achieved 57.7%, 69.0% and 61.9%. LLMs achieved 58.4%, 61.9% and 58.7%. We find that DeepEvidence retrieves evidence more comprehensively and correctly differentiates upstream and downstream relations among drug targets, pathways and biomarkers. This allows a more mechanistically grounded selection of surrogate endpoints.

DeepEvidence for EBM

EBM guides clinical practice by grounding decisions in high-quality clinical evidence²⁷. Central to EBM is the synthesis of published clinical studies, which must be updated as new findings emerge. Existing evidence can become outdated when subsequent randomized controlled trials alter the understanding of existing treatments, creating an evidence gap. To evaluate whether an AI agent can identify such gaps, we construct a benchmark focused on detecting newly relevant clinical trials. As illustrated in Fig. 5a, the agent receives the set-up of a systematic review, including the research question, review objectives and eligibility criteria, and must identify emerging trials that were not included in the original review but should be incorporated to maintain up-to-date clinical evidence.

We assess agent performance along two dimensions: (1) gap detection rate, defined as whether at least one true positive appears among the top 30 most plausible trials proposed by the agent, and (2) recall, defined as the proportion of true positives covered by the top thirty identified trials. Solving this task requires deep research into the literature and careful assessment of study eligibility using nuanced selection criteria.

The results are shown in Fig. 5d. DeepEvidence achieved the highest gap detection rate at 90.0%, compared with 50.0% for Biomni, 15.0% for ToolUniverse and 10.0% for PubMed-search LLMs. For recall, DeepEvidence reached 44.14%, again outperforming Biomni at 30.17%, ToolUniverse at 5.06% and LLMs at 2.67%. These results demonstrate

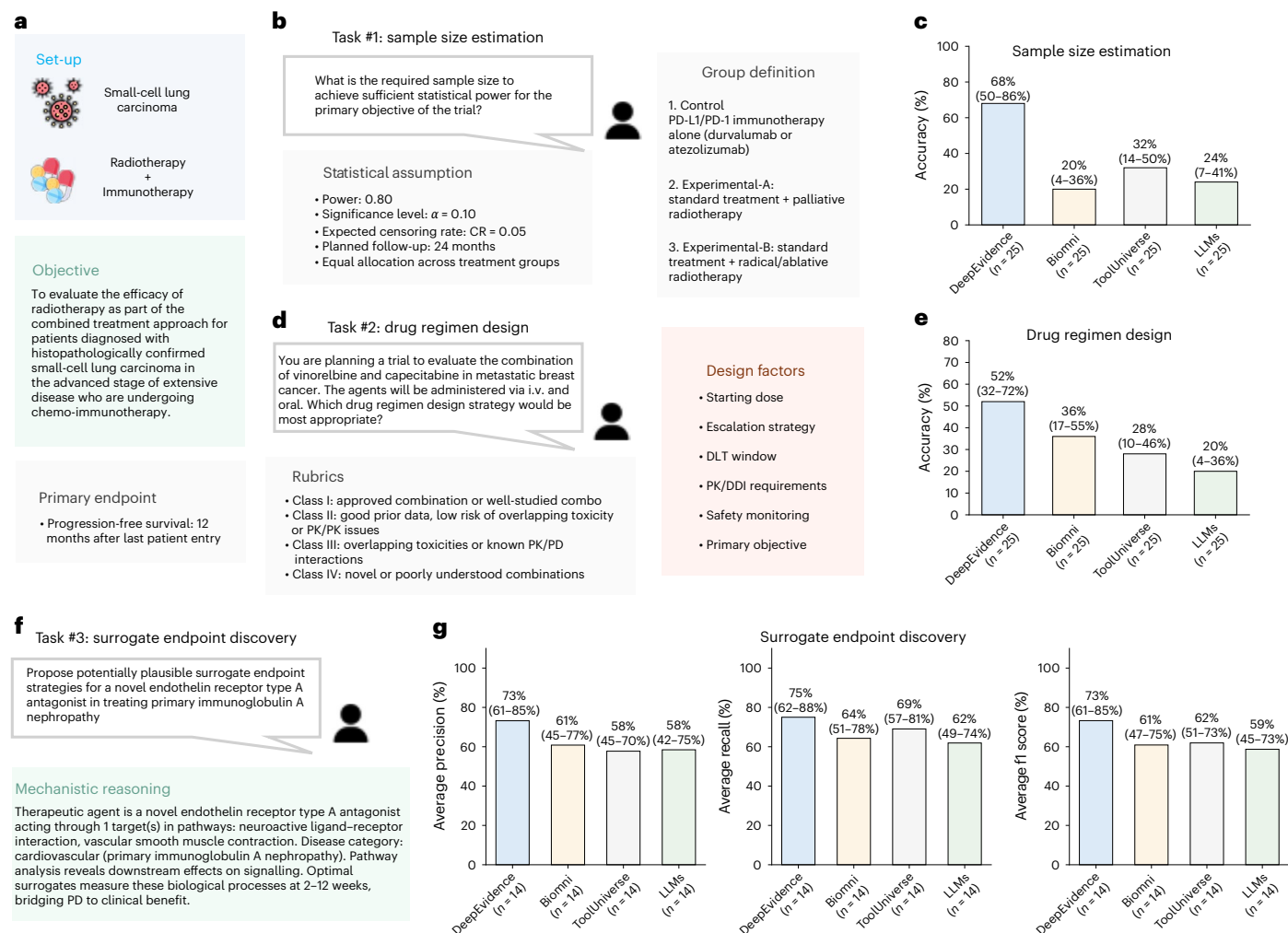


Fig. 4 | Experiment results of clinical trial development tasks. **a**, An example clinical trial set-up: assess the efficacy and safety of a combination treatment for lung cancer. **b**, Illustration of a sample size estimation task that specifies statistical assumptions and treatment group definitions for a multi-arm trial. **c**, Accuracy comparison for the sample size estimation task showing that DeepEvidence achieves substantially higher performance than baseline systems. All values are presented as mean (95% CI); the same format applies to all bar plots in this figure. PK, pharmacokinetics; PD, pharmacodynamics; DDI, drug–drug interaction. **d**, Illustration of a drug regimen design task that

defines rubrics for combination classes and outlines key design factors required to select an appropriate dosing strategy. **e**, Accuracy comparison for the drug regimen design task showing that DeepEvidence outperforms all baseline systems. **f**, Illustration of a surrogate endpoint discovery task where the agent uses mechanistic reasoning to link drug targets, pathways and disease biology to plausible surrogate measures. Bold values indicate that DeepEvidence outperforms the baseline. **g**, Performance comparison for the surrogate endpoint discovery task showing that DeepEvidence achieves the highest average precision, recall and F1 score among all methods.

that DeepEvidence is more effective in identifying evidence gaps and provides broader coverage of relevant studies. Its advantage comes from comprehensive deep research across related systematic reviews, randomized controlled trials and their citation networks, augmented by disease and drug knowledge graphs that enhance literature retrieval and study selection.

We further perform qualitative and quantitative analyses to better understand DeepEvidence's behaviour. Figure 5c presents an illustrative example of an evidence graph, showing how the framework traces mechanistic links among disease terms, treatment classes, mutation-specific populations and eligibility criteria to organize evidence and surface relevant supporting studies. Figure 5e further illustrates the iterative orchestration process, where the agent repeatedly performs deep research steps analogous to human reviewers conducting iterative evidence synthesis, while continuously retrieving and updating the evidence graph to track the research trajectory. In Fig. 5g, we provide a manual assessment of evidence graph quality based on 10 evidence graphs comprising 316 evaluated items (nodes and relations), auditing several dimensions: (1) provenance validity (whether nodes

correspond to real entities/sources), (2) entity/relation normalization correctness, (3) claim/observation alignment with source content, and (4) duplication rates. The results demonstrate high structural fidelity, with 100% provenance validity, 99.7% normalization accuracy and low duplication (0.6%). Relation-level correctness is also high ($\geq 99\%$). For claim alignment, which measures whether textual observations accurately reflect cited sources, we observe an overall accuracy of 93.3%, with most errors arising from nuanced interpretation of literature statements rather than incorrect entity linking.

To quantitatively evaluate the contribution of individual components, we conduct ablation studies and report results in Fig. 5f, comparing baseline variants and the full system. ReAct denotes a standard single-agent set-up following the ReAct loop, while ReAct+MCP represents the common practice of exposing all tool definitions directly in the model context for function calling. ReAct+MCP+CodeExec extends this with code execution, allowing the agent to inspect tool definitions and invoke tools programmatically. We further evaluate several multi-agent variants: Orchestrator+ReAct uses vanilla ReAct subagents without specialized search strategies, while BFRS-only and DFRS-only

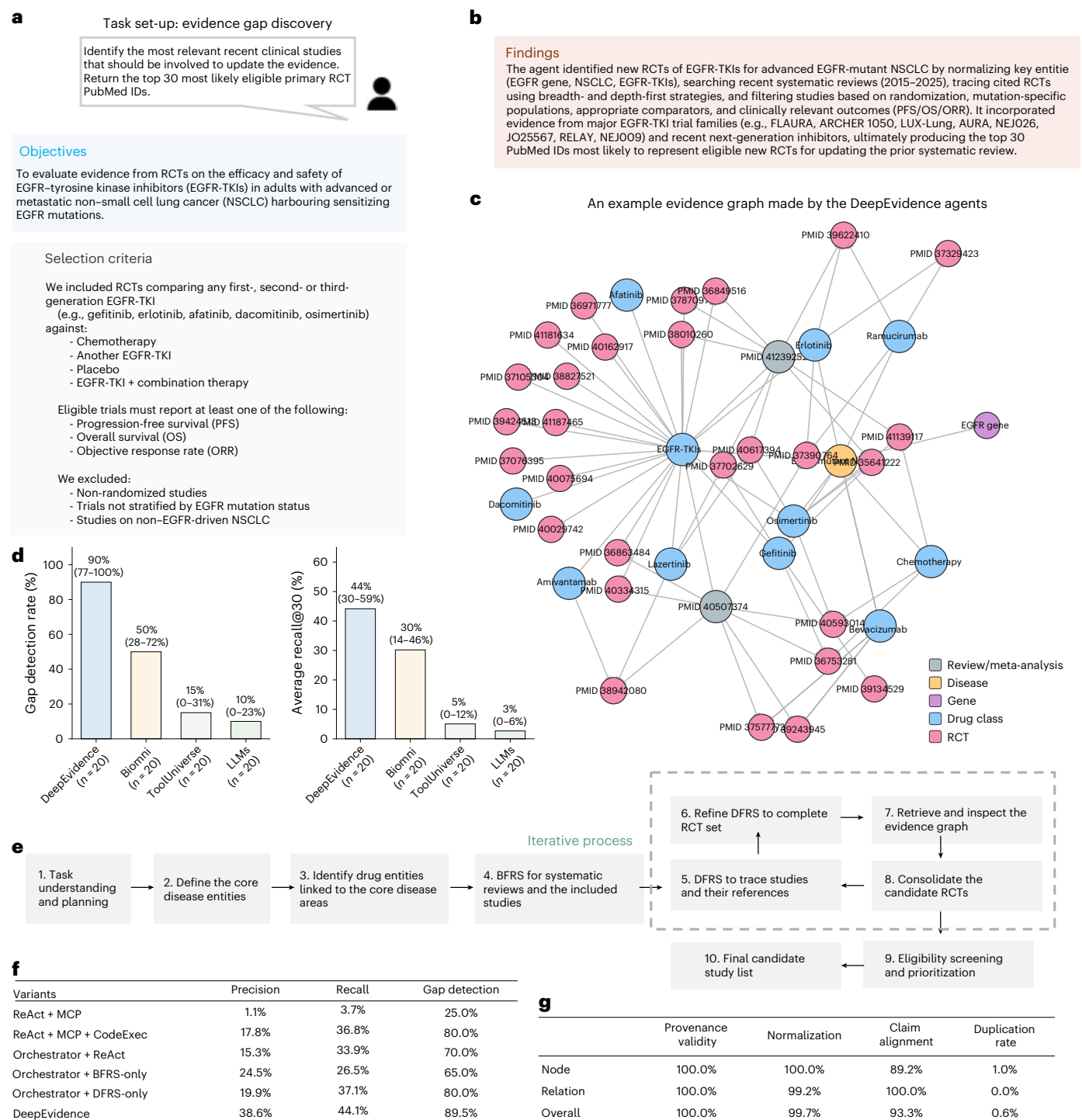


Fig. 5 | Experiment results of EBM tasks. a, Illustration of an evidence gap discovery task where the agent identifies recent randomized clinical trials (RCTs) that are most relevant for updating an existing clinical evidence. The input study objectives and detailed selection criteria are used to determine which trials qualify as eligible primary RCTs. **b**, An example finding made by DeepEvidence. **c**, An example evidence graph made by DeepEvidence. **d**, Performance comparison for the evidence gap discovery task showing that DeepEvidence achieves the highest gap detection rate and average recall at 30 among all methods. All values are presented as mean (95% CI); the same format applies to all bar plots in this figure. **e**, An example workflow for evidence gap discovery showing how the agent defines core disease and drug entities, performs BFRS and DFRS across studies, and iteratively consolidates and screens candidate RCTs.

The process ends with a prioritized list of eligible trials supported by an updated evidence graph. **f**, Ablation studies evaluating the contributions of key technical components in the proposed method. MCP: wrapping knowledge-base APIs as agent-accessible tool calls; CodeExec: enabling agents to write Python code for retrieving and processing data from knowledge-base APIs; Orchestrator+ReAct/BFRS/DFRS: multi-agent variants where subagents are instantiated as vanilla ReAct, BFRS-only or DFRS-only agents. **g**, Manual evaluation of the evidence graph quality. Provenance validity: whether nodes correspond to real entities supported by the cited literature; normalization: whether referenced entities are correctly normalized to canonical knowledge-base entities; claim alignment: whether recorded observations and relations faithfully reflect the cited source content; duplication rate: whether duplicate nodes or relations are present in the graph.

instantiate only one of the two proposed research strategies. The results reveal three key findings. First, directly exposing a large number of tools via MCP is suboptimal in our setting, particularly when many candidate tools are available, as agents are often confused by the excessive tool options; incorporating code execution substantially mitigates this issue. Second, multi-agent decomposition alone does not guarantee performance gains: all naive multi-agent variants underperform the single-agent ReAct+MCP+CodeExec baseline, indicating that subagent design is critical. In particular, a vanilla Orchestrator+ReAct set-up performs worse than the single-agent baseline across all dimensions, whereas introducing structured BFRS or DFRS strategies improves performance. Third, combining both BFRS and DFRS yields the best overall results, demonstrating that the two strategies provide complementary benefits and that their integration is critical to the full effectiveness of DeepEvidence.

Discussion

DeepEvidence is a deep research system for comprehensive investigation, synthesis and discovery across biomedical literature and knowledge bases. Importantly, the term ‘deep knowledge graph research’ does not imply that all information sources are explicit knowledge graphs, nor is DeepEvidence intended as another graph-based retrieval-augmented generation (GraphRAG) approach^{28,29}. Although some biomedical resources have a graph-like structure, such as KEGG with explicit nodes and relations, many important sources including PubMed and ClinicalTrials.gov are semi-structured or entirely unstructured. Existing graph-based retrieval and graph learning methods generally require preprocessing heterogeneous data into a unified knowledge graph before downstream modelling or traversal^{30,31}. Such preprocessing is difficult in biomedical settings because the underlying knowledge sources are fragmented, heterogeneous and continuously evolving. As a result, DeepEvidence should not be interpreted as a direct application of traditional BFS/DFS traversal over biomedical knowledge graphs³². Rather, it formulates LLM-driven agents as deep researchers operating under a graph-inspired exploration paradigm, where agents iteratively expand and refine evidence through BFRS and DFRS strategies while progressively constructing an evidence graph to organize findings, track provenance and support reasoning across heterogeneous knowledge sources.

Driving LLMs for deep research is a promising direction that has been increasingly explored across many domains^{18–20}. DeepEvidence aims to extend this paradigm to biomedical research, where several domain-specific challenges arise. First, biomedical deep research often requires long-horizon planning and multistep reasoning to comprehensively investigate available evidence without losing track of intermediate findings. This includes deciding which knowledge sources to query and how to coordinate among many heterogeneous tools and databases whose scope exceeds typical LLM context windows (for example, genes, diseases, drugs, pathways and trials). Second, biomedical reasoning must handle context specificity and reconcile conflicting evidence retrieved from different sources. Third, the research process must remain transparent so that intermediate evidence and conclusions can be validated by human experts. These challenges motivate the design of DeepEvidence. Specifically, we introduce a multi-agent framework in which an orchestrator with a clean and focused context monitors overall research progress and spawns specialized subagents for targeted exploration. We leverage the parametric knowledge of powerful LLMs to interpret context-dependent evidence and reconcile heterogeneous findings, which is often difficult for purely graph-based learning methods trained from scratch. We further construct an explicit evidence graph to organize retrieved evidence, preserve provenance and support transparent reasoning. Together, these design choices form a principled framework for biomedical deep research and are empirically validated across a broad range of tasks in drug discovery, preclinical research, clinical development and EBM.

Several limitations remain. Although we have incorporated various biomedical entities and domain-specific knowledge sources, coverage is still incomplete, and the current benchmarks reflect only a subset of biomedical research tasks. In addition, the system currently works with well-curated knowledge bases such as KEGG and Gene Ontology that offer stable schemas and high-quality APIs. Future versions must adapt to proprietary or internally hosted sources that may contain missing values, inconsistent structure or incomplete interfaces. This requires empowering the agent to construct new tools for accessing these sources. Another limitation lies in handling context specificity and conflicting evidence during the research process. While DeepEvidence can retrieve and preserve heterogeneous evidence with provenance, it currently lacks explicit probabilistic modelling or uncertainty-aware mechanisms to formally reconcile conflicting findings across sources, and instead relies primarily on the parametric reasoning ability of the underlying LLM to interpret context and weigh evidence. Future work may incorporate probabilistic reasoning frameworks or calibrated confidence modelling to improve robustness in conflict-heavy settings. Moreover, the present system focuses on two research strategies: BFRS and DFRS. These strategies substantially improve over generic agents, but they are unlikely to be optimal. DeepEvidence has not yet reached expert-level performance on many tasks, and future improvements will depend on both stronger underlying language models and methods that allow the agent to discover improved research strategies through self-directed adaptation. Finally, our current work focuses mainly on text-based modalities, whereas we could further extend it to multimodal data such as imaging, structural biology and proteomic information to make it more applicable to broader biomedical research tasks.

Looking forward, we envision DeepEvidence as a foundation for deeper exploration of literature and knowledge bases in biomedicine and as a complementary component to general-purpose AI agents. Through closer collaboration with human experts, DeepEvidence can support research that requires rigorous evidence grounding and can assist in generating and validating scientifically meaningful hypotheses through a structured and transparent deep research paradigm.

Methods

Processing open benchmarks

The open benchmark experiments are constructed by selecting and adapting tasks from HLE²¹, LabBench-LitQA2²², SuperGPQA²³ and TrialPanorama²⁴. Our evaluation design is guided by the intended use case of DeepEvidence: literature-driven, multisource biomedical research. Accordingly, we prioritize tasks that require deep literature exploration, cross-source evidence synthesis and non-trivial reasoning, rather than those solvable via shallow retrieval and standard retrieval-augmented generation.

For HLE, we first select questions whose subject area is labelled as ‘Medicine’. We then remove questions that require an additional image as input. This filtering step is motivated by our observation that medical questions in HLE often require literature-grounded reasoning for diagnosis or clinical decision-making, rather than visual interpretation. After filtering, this process yields a total of 30 questions.

For LabBench, we focus on the LitQA2 subset that has 184 questions, as this component explicitly requires exploration of scientific literature to support reasoning and answer generation, making it well aligned with our target setting. For SuperGPQA, we first restrict the benchmark to questions with difficulty level labelled as ‘Hard’. Within this subset, we further select questions whose field is ‘Clinical Medicine’. This results in a curated dataset containing 172 questions, which we refer to as the SuperGPQA hard clinical medicine set. This targeted selection emphasizes challenging, domain-relevant questions, as easier questions can often be addressed by parametric knowledge or simple retrieval and are therefore less informative for evaluating deep-research capabilities.

For TrialPanorama, we select the ‘EvidenceSummary’ task from the benchmark. We then retain the most recent 50 questions. In the original formulation, each question is an evidence-summarization task that presents abstracts from multiple studies and asks the model to summarize the evidence to select the correct option. In our setting, we remove these abstracts from the input context, thereby requiring agents to explore the literature to identify the correct answer actively. This modified set-up forms our TrialPanorama-EvidenceQA dataset.

Curating target identification tasks

For the target identification task, we aim to evaluate an agent’s ability to identify therapeutically actionable targets through mechanistic reasoning and multisource evidence synthesis, rather than parametric recall. Accordingly, we intentionally avoid using clinically validated targets as ground truth, as such targets are often memorized in LLM pretraining corpora and would allow models to answer via direct recall. Instead, we construct a structured proxy task where agents must prioritize candidates under a given pathway mechanism. While this ground truth does not represent clinical truth, it provides a controlled and interpretable signal for evaluating evidence-consistent reasoning.

The evaluation instances are made by extracting pathway topology from KEGG disease pathway files in KGML format. We parse 85 human disease pathways spanning cancer, metabolic disorders, infections and autoimmune conditions to build directed signalling graphs where nodes represent genes/proteins and edges encode regulatory relationships (activation, inhibition, repression or expression). For each disease pathway, we formulate target identification as a multiple-choice question asking which genes represent the most promising therapeutic targets, with ten candidate options drawn from pathway members. An example instance of this task is presented in Supplementary Table 2. Although KEGG is used for benchmark construction, it provides only partial, local pathway structure and does not directly yield the correct answers. Solving the task requires integrating heterogeneous evidence (for example, literature, druggability and pathway context), rather than performing a single-source lookup. All compared methods have access to similar knowledge sources and are not given privileged information about the benchmark construction, ensuring a fair comparison.

A topological logic engine is developed to assign gain scores (0, 1 or 2) to each candidate target based on three criteria. First, we compute path polarity by finding all simple paths from a candidate gene to disease endpoint nodes and calculating the cumulative product of edge weights along each path, where activation edges contribute +1 and inhibition edges contribute −1. Genes with average path polarity >0.5 (that is, promoting disease progression) receive gain 2 when targeted with an inhibitor, while genes with negative polarity (suppressing disease) receive gain 0 to reflect the biological contradiction of inhibiting a protective factor. Second, we incorporate betweenness centrality from graph-theoretic analysis: genes in the top 10% of centrality scores that also show positive path polarity are prioritized as optimal targets (gain 2), as they represent critical bottlenecks for signal convergence. Third, we apply druggability filtering by assigning gain 0 to non-druggable protein classes (for example, ribosomal proteins and cytoskeletal components) identified via keyword matching against gene symbols.

We further implement disease-specific logic profiles that weight gene functional classes differently across pathway types. For cancer pathways, kinases and enzymes receive elevated priority; for drug resistance pathways, transporters and receptors are emphasized; for infection pathways, pattern recognition receptors and cytokines are prioritized. Gene functional types are inferred from KGML entry metadata, including enzyme commission numbers for enzymes and graphics annotations, with fallback to symbol-based pattern matching against curated gene family dictionaries. The final multiple-choice format presents options as `GENE_SYMBOL – functional_type` with answers corresponding to all gain 2 options, enabling evaluation of whether an AI agent can identify therapeutically actionable targets from pathway topology.

Curating target validation tasks

We introduce an evaluation task termed target validation, consisting of 30 manually curated questions. To construct this dataset, we manually select target candidates that exhibit conflicting evidence in the literature, such that answering each question requires aggregation and reconciliation of evidence from multiple sources rather than reliance on a single supporting study. These conflicts generally fall into three categories: (1) evidence hierarchy conflict, where strong mechanistic or preclinical evidence supports a target but higher-level clinical evidence (for example, failed clinical trials) contradicts its therapeutic potential; (2) context or applicability conflict, where findings are valid in general biological settings or specific diseases/subtypes reported in the literature but are not applicable to the context defined in the question; and (3) evidence noise and temporal inconsistency, where evidence is mixed or evolves over time, such as early-phase studies showing positive signals while later-phase trials fail, or conflicting conclusions reported across studies. Due to the presence of these conflicts, agents answering the questions will naturally retrieve conflicting evidence during the research process and must reconcile these discrepancies to arrive at a well-supported final judgement. This dataset therefore enables systematic evaluation of how DeepEvidence integrates multisource evidence and resolves conflicts under realistic biomedical research settings.

Curating in vivo metabolic flux response tasks

For the in vivo metabolic flux response prediction task, we design a controlled setting to evaluate whether an AI agent can perform multi-step reasoning over metabolic pathway structure to infer flux perturbations following enzyme inhibition. Rather than reproducing true in vivo outcomes, we focus on mechanistic reasoning under structured constraints and avoid using real experimental ground truth, which may be memorized by LLMs and enable parametric recall. Instead, we derive ground-truth signals from pathway topology as an interpretable proxy, acknowledging that this abstraction simplifies biological complexity but provides a controlled testbed for deep reasoning. KEGG is used solely for benchmark construction and does not directly yield answers. Solving the task, therefore, requires multihop reasoning beyond single-source lookup.

Concretely, we parse KEGG disease pathway files in KGML format, extracting both the reaction graph (substrate → product relationships) and the signalling graph (regulatory relationships) to model metabolic flux dynamics. For each pathway, we select a target gene and formulate a multiple-choice question asking which tumour cohorts would exhibit optimal metabolic flux responses upon target inhibition, with seven candidate phenotypic outcomes. An example instance of this task is presented in Supplementary Table 4.

Ground-truth answers are computed from pathway topology using a three-layer verification system. First, we apply a topological consistency check: using breadth-first search from the target gene node, we identify downstream metabolites (products reachable within two reaction steps) and upstream substrates (precursors within two steps). By the logic of enzyme inhibition, downstream products should show reduced isotope tracer labelling, while upstream substrates should accumulate. Options reflecting this pattern receive gain 2, whereas options stating the opposite (for example, ‘increased downstream product levels despite inhibition’) receive gain 0, as they contradict mass-balance principles. Second, we identify pathway endpoints as terminal nodes with zero out-degree in the reaction graph and mark sustained suppression of endpoint synthesis as an optimal response indicator. Third, we detect feedback loops via strongly connected component analysis; nodes participating in cycles may exhibit transient or compensatory responses, receiving gain 1 to reflect partial but unstable target engagement.

To ensure biological grounding, the option text is dynamically generated using actual compound names extracted from the KGML

reaction elements rather than generic placeholders. For each target gene, we traverse the reaction graph to collect real metabolite names (for example, 'Tumours showing a 40% decrease in pyruvate labelling' rather than abstract references), providing concrete molecular context. The metabolic dependency and process descriptors in question stems are similarly derived from pathway-specific compounds when available, falling back to curated templates (for example, 'glutamine metabolism' and 'redox balance') only when compound extraction fails. The final multiple-choice format presents seven phenotypic outcomes with shuffled order, where answers correspond to all gain 2 options, enabling evaluation of whether an AI agent can reason about metabolic flux perturbations from pathway topology.

Curating mechanism of action and pathway reasoning tasks

For preclinical research tasks spanning oncology and molecular biology, we curated evidence-informed multiple-choice questions comprising both single-select and multiple-select items. These questions are designed to reflect realistic preclinical research scenarios in which correct conclusions depend on mechanistic specificity, experimental context and methodological constraints, rather than isolated factual recall. We defined the scope of preclinical research tasks by identifying recurrent analytical and decision-making problems encountered in preclinical studies. The curated questions cover core themes, including targeted drug mechanism of action characterization, limitations of tumour therapeutic strategies, cancer cell line properties and drug sensitivity, omics data analysis workflows, signal transduction mechanisms, experimental assay validation, tumour molecular subtypes and metastasis mechanisms, and associations between gene expression patterns and phenotypic or clinical readouts. Topics were preferentially selected from areas with preclinical consensus to minimize reliance on preliminary, controversial or contradictory findings.

For each task, question stems and candidate answers were grounded in authoritative preclinical evidence sources selected according to task relevance. These sources include peer-reviewed preclinical literature describing *in vitro* experiments, *in vivo* tumour models and molecular mechanism studies, as well as curated resources commonly used in preclinical research, such as cancer cell line repositories (for example, ATCC and DSMZ), drug sensitivity databases (for example, GDSC), gene expression and survival analysis platforms (for example, GEPIA2, GEPIA3, TIMER and Kaplan–Meier Plotter) and pathway databases (for example, KEGG). Established experimental principles and technical guidelines were also used where appropriate. Evidence sources were applied at the level of individual tasks to ensure internal consistency and traceability of claims, without enforcing a uniform or automated validation pipeline across all items. Each task was formulated as either a single select or multiple select question, depending on whether the underlying preclinical problem admits a unique optimal answer or multiple concurrently valid interpretations.

Answer options were manually curated by domain experts and assigned graded correctness labels, with gain equal to two, one or zero, reflecting increasing deviation from task consistent preclinical evidence. A strictly classified gain 2 option must simultaneously align with definitive preclinical evidence or widely accepted mechanistic understanding, be supported by peer-reviewed literature or curated database records without contradictory findings, explicitly match the experimental scenario implied by the question, including precise specification of molecular targets, cell line characteristics, assay methods or data interpretation standards, and ensure that all biological claims are factually exact, unambiguous and traceable to specific evidence such as literature figures, tables, database records or experimental protocols. Gain 1 options represent partially correct reasoning paths that incorporate valid biological or methodological elements but omit critical constraints, overgeneralize from related systems or rely on associations insufficient to support a definitive conclusion. Gain 0 distractors were constructed to be mechanistically

plausible yet inconsistent with established evidence or experimental logic, often arising from cross-entity conflation, misinterpretation of assay capabilities, or direct contradiction of consensus findings, and, in some cases, adapted from true statements in adjacent research contexts to preserve surface plausibility. All items were reviewed to ensure that answer correctness cannot be resolved using generic biological knowledge alone, but instead requires reasoning grounded in the specific experimental or analytical context of each task, and that no explicit database accession numbers or record identifiers appear in the question text, while references to commonly used platforms are retained where methodologically relevant.

Curating clinical trial development tasks

For the sample size estimation task, we start from the sample-size estimation instances provided in the TrialPanorama benchmark. We then construct our evaluation split by selecting the 25 most recent trials in that collection as the test set. For each instance, we treat the benchmark-provided ground-truth sample size as the correct answer and convert the problem into a multiple-choice question by generating four plausible distractors via multiplicative perturbations of the true value. Specifically, we sample a perturbation ratio in the range 0.25–4.00 (that is, 75% smaller to 300% larger than the ground truth), apply it to the true sample size, round to an integer and enforce basic validity constraints (for example, within a reasonable sample-size range and not equal to the correct value). We then shuffle the five options and record the corresponding option letter as the answer label, enabling straightforward accuracy-based evaluation of whether an AI agent can select the correct sample size from competing alternatives. An example instance of this task is presented in Supplementary Table 6.

For the drug regimen design task, we constructed an evidence-guided multiple-choice dataset from a structured toxicity and dose-finding corpus that was manually extracted from early-phase oncology combination trials³³. We first loaded trial-level metadata (for example, population and DLT evaluation window), regimen-level drug attributes (agent names and administration routes), dose-level ladders, observed DLT summaries by dose level, protocol DLT definitions, reported maximum tolerable dose (MTDs) and escalation-design descriptors (when available) from normalized tables. Using all single-agent regimens in the corpus, we computed monotherapy baselines for each drug, including a reference monotherapy MTD (maximum reported MTD across monotherapy trials) and a deduplicated list of monotherapy DLT terms. For each multi-agent regimen, we then derived regimen-specific evidence features by (1) analysing the dose ladder to estimate starting and maximum dose intensities relative to monotherapy MTDs, (2) quantifying DLT patterns across dose levels and estimating a toxicity-overlap score as the fraction of combination DLT types that were also observed in monotherapy, and (3) extracting auxiliary context such as population and route of administration. These evidence features were mapped to a four-level design-class label (class I–IV) using a rule-based classifier that prioritizes approval status proxies, availability of monotherapy dose-finding evidence, toxicity overlap and interaction risk proxies. Finally, for each eligible regimen we generated a design-strategy multiple-choice question: the question stem summarizes the supporting evidence, and all answer choices use the same drug combination but differ only in the proposed trial design strategy corresponding to each class (for example, safety verification/limited adjustment, de-escalation near monotherapy MTD, standard escalation or cautious lead-in with extended DLT window and mandatory PK/interaction assessment). We excluded monotherapy regimens and combinations with insufficient dose/DTL evidence. An example instance of this task is presented in Supplementary Table 7.

For the surrogate endpoint discovery task, we curate a set of recently approved drugs from KEGG (drug entries with approval dates in 2025) as starting points, and automatically construct mechanistically grounded multiple-choice questions from their pathway annotations.

For each drug, we retrieve a structured drug record from KEGG (name, clinical comment/efficacy text, disease annotations, class labels, target gene identifiers and linked pathways) and keep drugs with at least one mapped target and pathway. We then download KGML files for up to five associated pathways and parse them into directed interaction graphs. To connect drug mechanism to measurable intermediate outcomes, we perform a bidirectional breadth-first traversal from the drug's target nodes in the merged pathway graph (depth limit 10) and match reachable genes to a predefined library of biological-process marker sets (for example, proliferation, apoptosis, metabolism, DNA damage, inflammation, angiogenesis, migration and signalling). We also categorize each drug's therapeutic context (for example, cancer, inflammation, metabolic, cardiovascular, neurology or other) using keyword heuristics over high-priority clinical text fields (COMMENT/EFFICACY), complemented by DISEASE and CLASS fields. Using the inferred downstream process effects, we generate drug-specific, disease-aware surrogate endpoint strategies as the correct options (gain 2), emphasizing distal biomarkers measurable within 2–12 weeks that bridge pharmacodynamic modulation to expected clinical benefit; we additionally include acceptable but weaker strategies (gain 1) that are poorly timed or weakly linked to clinical outcomes. To create challenging distractors (gain 0), we adopt a two-pass construction: we first generate correct surrogate strategies for each drug, then, for a given drug, we sample distractors from (1) correct surrogate strategies generated for other drugs (mechanistically plausible but mismatched to the current mechanism/indication) and (2) upstream/proximal measurements (depth <3 from target) that primarily reflect target engagement rather than downstream benefit. Each final item is a multi-answer MCQ with six to ten options, where all gain 2 options are labelled as correct, and we enforce quality constraints such as requiring multiple correct/suboptimal/distractor options and removing explicit database identifiers (for example, KEGG IDs) from question text. An example instance of this task is presented in Supplementary Table 5.

Curating EBM tasks

For the EBM task, we construct the test data from versioned Cochrane systematic reviews whose conclusions changed between consecutive updates. Starting from a curated list of updated reviews, we programmatically retrieve both the most recent version and its immediately preceding version by resolving digital object identifiers (DOIs) to PubMed records and downloading full XML via the National Center for Biotechnology Information (NCBI) E-utilities. From each review version, we extract structured metadata (title and structured abstract sections) and parse the reference lists explicitly labelled by the review (for example, references to studies included in this review and references to studies excluded from this review). This yields a version-resolved mapping from each review to the set of included study PMIDs, enabling robust pairing of review versions using a shared base DOI and precise comparison of included-study sets. An example instance of this task is presented in Supplementary Table 8.

We then build the final EBM test set as an evidence-gap identification task. For each paired review (older versus newer), we provide the updated review's research context (for example, objectives, eligibility/selection criteria and outcomes) and, when available, the list of studies included in the prior version. The model is asked to identify which new studies should be added when updating the review. Ground-truth answers are defined as the set difference between the newer and older versions' included-study PMIDs, that is, studies that appear in the updated review's included list but were not included previously.

Curating open-ended research challenges

While multiple-choice questions settings enable standardized and quantitative evaluation (for example, accuracy) and facilitate fair comparison across methods, they simplify the open-ended nature of real-world biomedical discovery. To better reflect practical usage,

we complement our benchmark suite with two open-ended research challenges that require free-form, end-to-end reasoning without predefined answer options. These tasks are designed to evaluate an agent's ability to generate hypotheses, synthesize multisource evidence and make scientifically grounded decisions under realistic conditions.

The first task is target prioritization. This task focuses on validating and prioritizing candidate therapeutic targets derived from data-driven discovery processes (for example, RNA sequencing differential expression analysis). Agents are provided with a set of candidate genes and asked to design a prioritization strategy, identify high-confidence targets and justify their decisions. The task requires distinguishing likely causal drivers from correlated signals, assessing druggability, proposing therapeutic strategies, considering safety implications and outlining validation approaches. Importantly, agents must explicitly differentiate conclusions supported by prior evidence from informed hypotheses. An example instance is presented in Supplementary Table 9. Outputs are evaluated by domain experts using structured rubrics that assess multiple dimensions, including causal reasoning, mechanistic specificity, prioritization quality, therapeutic strategy design, druggability assessment, evidence awareness and systems-level reasoning. The full rubrics are presented in Supplementary Table 11.

The second task is translational validation. This task evaluates whether a candidate intervention is sufficiently supported by pre-clinical evidence to justify advancement towards clinical development. Agents are asked to synthesize mechanistic, *in vitro* and *in vivo* evidence, reconcile potentially conflicting findings and provide a justified decision on clinical progression. The task emphasizes deep, multistep reasoning across disease mechanisms, target biology and therapeutic effects. An example instance is presented in Supplementary Table 10. Outputs are evaluated by domain experts using structured rubrics that assess: (1) biological correctness and logical consistency, (2) relevance to the question, (3) depth of causal reasoning (for example, disease → target → pathway → phenotype chains), (4) mechanistic specificity, (5) evidence handling (distinguishing direct versus indirect evidence), (6) hallucination control, (7) uncertainty calibration and (8) answer specificity. These rubrics are designed to capture both the correctness and scientific rigour of the generated responses. The full rubrics are presented in Supplementary Table 12.

Implementing the orchestrator and research agents

DeepEvidence employs a hierarchical multi-agent architecture implemented using LangGraph, an open-source framework for building agent and multi-agent workflows through explicit state-machine graphs. At the top level, an orchestrator agent receives user research queries and dynamically plans the investigation strategy by decomposing complex questions into targeted subtasks. The orchestrator prompt is available in Supplementary Note A.1. The orchestrator maintains a state machine that tracks search budgets, manages the evidence graph and coordinates two specialized research subagents: the BFRS subagent and the DFRS subagent. BFRS is designed to explore broadly across specified knowledge bases, gathering diverse information about multiple entities simultaneously, while DFRS follows citation chains and entity relationships in depth to uncover detailed mechanistic evidence. The BFRS and DFRS prompts are presented in Supplementary Notes A.2 and A.3, respectively. The orchestrator invokes these subagents through tool calls that specify the search target (including entity names and standardized identifiers such as PubTator IDs or Unified Medical Language System (UMLS) Concept Unique Identifiers (CUIs)), the target knowledge bases and the action budget allocated for the search. Each subagent operates as an independent state graph with its own tool execution nodes, allowing modular composition and controlled resource allocation.

All agents utilize code execution as their primary action mechanism. Rather than returning static outputs, each tool generates

executable Python code that is run within a sandbox. This approach enables agents to perform multistep API queries, data transformation and statistical analysis. For example, when searching for genes, the agent executes code that queries multiple databases, aggregates the results into a unified dataframe, and saves the outputs to the shared workspace for subsequent analysis steps. Execution outputs, including runtime errors and failed API calls, are returned to the agent as part of the context, enabling iterative self-correction through the ReAct loop where agents inspect error messages, revise code and retry execution. To ensure robustness, the environment enforces action budgets that limit the number of execution steps and prevent infinite loops or excessive resource usage. Large-scale data and potential memory issues are handled through incremental processing strategies such as slicing, filtering and partial loading, while timeout or rate-limit failures trigger bounded retries within the allocated action budget. Although full reproducibility is inherently limited by LLM non-determinism and evolving external APIs, the code execution process itself remains deterministic given fixed inputs and API states. In addition, the evidence graph records intermediate retrieved entities, relations and execution outputs, providing a transparent and traceable reasoning trajectory for downstream inspection and reproduction.

DeepEvidence implements an explicit evidence graph as a first-class memory structure that the agent incrementally constructs and maintains throughout the deep research process. The graph stores only high-value normalized biomedical entities such as genes, diseases, drugs, pathways, papers and concrete findings, each identified by canonical identifiers when available and deduplicated globally. Edges represent a controlled set of mechanistic membership association and evidence level relations, and every node and relation is grounded with explicit provenance from primary literature or curated knowledge graphs. Rather than encoding context as free text, the agent attaches short factual observations to entities, capturing experimental setting methodology and quantitative or mechanistic outcomes. During reasoning, the agent queries the existing graph before adding new content, merges near duplicates, updates observations instead of creating redundant nodes, and records conflicting findings with separate evidence links. We add the prompt in Supplementary Note A.4 to the agent's system prompt as additional guidance. This tight coupling between the agent and the evidence graph enables iterative hypothesis refinement, evidence aggregation and transparent traceability, ensuring that conclusions are derived from an explicit, structured and auditable body of biomedical evidence.

To facilitate cross-knowledge-base research and entity bridging, we implemented unified modality-wise search tools that aggregate information from multiple authoritative sources through a single interface. For instance, the unified gene search tool queries BioThings (MyGene.info), KEGG and Open Targets simultaneously, returning consolidated results with cross-database identifiers. Similar unified tools exist for diseases, drugs, compounds, targets, pathways and variants. Beyond entity lookup, we integrated relation-based search capabilities via the PubTator3 API, enabling agents to discover entities linked through specific semantic relationships such as TREAT, CAUSE, INTERACT, INHIBIT and ASSOCIATE. This allows agents to traverse from diseases to candidate drugs, from genes to associated phenotypes, or from chemicals to their biological targets, supporting the depth-first reasoning patterns essential for mechanistic hypothesis generation. Notably, DeepEvidence does not construct unified knowledge graphs on the fly; instead, these tools issue targeted queries to heterogeneous external sources in parallel, with retrieved results stored in the sandbox for downstream processing and analysis. As queries are executed concurrently, retrieval latency is primarily bounded by the slowest responding API rather than scaling linearly with the number of queried sources or the size of the underlying knowledge bases. In practice, total runtime depends on both external API latency and downstream processing complexity, such as filtering, joining and multihop reasoning.

Implementing the knowledge graph tools

We included the following knowledge bases in the tool library for the agent: BioThings⁶, ChEMBL²⁵, ClinicalTrials.Gov³⁴, Gene Ontology⁴, Human Phenotype Ontology³⁵, KEGG³, NCBI Datasets API³⁶, OpenFDA²⁶, Open Genes³⁷, ProteinAtlas³⁸, PubChem³⁹, PubTator¹, PubMed³⁶, Reactome⁴⁰, UMLS⁴¹ and UniProt⁴². The detailed tool list is presented in Supplementary Table 1.

We developed a modular tool library to enable agents to access and query 17 biomedical knowledge graphs and databases. Each knowledge graph integration follows a consistent two-layer architecture: (1) a low-level client class that encapsulates API authentication, rate limiting, error handling and response parsing; and (2) high-level tool functions that provide task-oriented interfaces returning pandas DataFrames with formatted output strings suitable for agent consumption. This separation allows the same underlying API client to support multiple tool functions while maintaining code reusability. All tool functions follow a uniform signature pattern, accepting search parameters and an optional `save_path` argument, and returning a tuple of (DataFrame, formatted_output_string). The formatted outputs are designed to be both human-readable and parseable by LLMs, including summary statistics, result previews and file save confirmations.

For RESTful APIs (BioThings, PubChem, KEGG, ClinicalTrials.gov, OpenFDA, ProteinAtlas, Open Genes, NCBI Datasets and HPO), we implemented HTTP-based clients using the `requests` library with configurable timeouts, retry strategies and custom user-agent headers. These clients parse JSON or flat-file responses into structured dictionaries. For GraphQL-based APIs (Open Targets Platform), we constructed query templates with parameterized variables for target, disease and drug searches. For programmatic APIs requiring authentication (UMLS and UniProt), we implemented credential management and session-based authentication flows. PubTator3 integration includes specialized support for entity-typed boolean queries (for example, `@CHEMICAL_remdesivir AND @DISEASE_COVID_19`) and relation-based searches using semantic predicates (TREAT, CAUSE, INTERACT, INHIBIT and ASSOCIATE). Some knowledge graphs required special handling: ChEMBL supports SMILES-based similarity and substructure searches using Tanimoto coefficients; Reactome uses stable identifiers for pathway hierarchy traversal; and KEGG employs a unique flat-file format requiring custom parsing logic.

To enable cross-database entity resolution and comprehensive information retrieval, we implemented unified search tools that aggregate results from multiple sources through a single interface. For example, `search_genes_unified` concurrently queries BioThings (MyGene.info), KEGG, Open Targets and optionally MyVariant.info, merging results with cross-referenced identifiers (Entrez, Ensembl and KEGG IDs). Similarly, `search_drugs_unified` aggregates from BioThings (MyChem.info), OpenFDA (Drugs@FDA and drug labelling), KEGG Drug, Open Targets and ChEMBL. These unified tools use concurrent execution via `ThreadPoolExecutor` to minimize latency and return consolidated DataFrames with source attribution. The tool outputs are designed to support downstream analysis: JSON files for detailed records, CSV files for tabular data, and markdown-formatted summaries for immediate interpretation by the research agents.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The test datasets used in the experiments are publicly available at <https://huggingface.co/datasets/zifeng-ai/DeepEvidence>. Source data are provided with this paper.

Code availability

The code implementation of DeepEvidence is available via GitHub at <https://github.com/RyanWangZf/BioDSA/tree/main/biodsa/agents/deepevidence> (ref. 43).

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Author contributions

Z.W. and J.S. conceived the study and led the writing of the manuscript. Z.W. developed the framework, designed the methodology, conducted the primary experiments, analysed the results and led the overall project. Z.C., Z.Y. and X.W. contributed to experimental evaluation, benchmark construction and data collection. Q.J., Y.P. and Z.L. contributed to scientific discussion, manuscript revision and interpretation of the biomedical applications and results. All authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Zifeng Wang or Jimeng Sun.

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Sample size

No statistical methods were used to predetermine sample size. Sample sizes were determined by data acquisition followed by task-specific filtering and curation pipelines described in Methods. For the open biomedical benchmarks, we evaluated DeepEvidence on 30 questions from Humanity's Last Exam Medicine, 184 questions from LabBench LitQA2, 172 questions from SuperGPQA Hard Clinical Medicine, and 50 questions from TrialPanorama EvidenceQA. For the curated biomedical-discovery benchmarks, sample sizes were 25 for target identification, 30 for target validation, 25 for mechanism-of-action and pathway reasoning, 25 for in vivo metabolic flux response prediction, 25 for sample size estimation, 25 for drug regimen design, 14 for surrogate endpoint discovery, and 20 paired systematic reviews for evidence gap discovery. Evidence graph quality was audited on 10 evidence graphs comprising 316 evaluated items. The breadth of the evaluation suite yields statistically significant results across diverse biomedical research tasks.

Data exclusions

We removed the benchmark data for the tasks have excessive numbers of samples.

Replication

n/a. Replications are not applicable because the computational results are deterministic without randomness.

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